

Part I – Foundations

All variants of chess, in particular the 3D variants, have one thing in common; they show respect for the invariants of 2D chess. However, in the excursion to three dimensions, some of these invariants display a duality, they split into incompatible patterns; one must be chosen at the expense of another. Such invariant pairs are referred to as *degenerate rules*: different ways of specifying a single rule that have exactly the same consequences in one realm but divergent consequences in a higher abstraction. Which invariants are *honored*, and which *shunned*, pretty much determine the rules.

In most attacks on the problem of playing chess in three dimensions, the canonical invariant is that pieces which advance move in *straight lines*. In the approach advocated here, the canonical invariant is that pieces which advance move in *one dimension less* than the board.

In 2D this means that pieces advance, coincidentally, in straight lines ($2 - 1 = 1$). The two rules, “advance in a straight line” and “advance in a space one dimension less than the board” are degenerate in 2D. This degeneracy is broken in 3D.

In 3D, advancing in a space one dimension less than the board means pieces advance in flat planes ($3 - 1 = 2$). These two different ways of expressing the “advance” rule scale to higher dimensions in very different ways, with huge consequences.

Planar moves are the initial breakthrough that yields hope there is in fact *one right rule set* that leads to play that genuinely *feels like chess*. They are the basis for how pieces *should* move.

The objective of Part I is to discover the correct *size and shape* of the board, the right *set of pieces*, and the particular *planes* each piece should move in.

“The mind, once stretched by a *new idea*, never returns to its original dimensions.”

— Ralph Waldo Emerson

1. The Trajectory Paradigm

While the moves for each piece will be presented individually in Part II, they share a common heritage which is best illuminated by considering the rook. A robust paradigm barrier separates chess in two dimensions from chess in higher dimensions. A mathematical approach to discovering the rules to 3D chess would encounter this paradigm somewhat later in the development. However, there is some tedium involved in that path, so instead we'll head to high ground for a preliminary view of the promised land. The intent is to convince the reader that something new and exciting has been discovered, something unique that sets this rule set apart from others, a paradigm breakthrough. The breakthrough is what validates the claim of *one right rule set*, and justifies the intellectual investment required to follow a formal line of development.

Of necessity, this means that some initial, plausible choices about the fundamentals of 3D chess, such as board size and shape, will be temporarily assumed without justification. For instance, the 3D board shall be regarded as a cube of cubes, and the dimension agnostic term *tile* will be used to refer to either square or cube, trusting in context to disambiguate. Even the piece designs will have a justification. Said justifications will be more rigorously developed in Part II as the design of each piece derives from the explicit rule for how it moves. It is a long-standing chess tradition that the design of each piece alludes to how it moves; one more pillar in the challenge of finding a rule set that *feels like chess*.

In due course, a formal color-coding scheme will be introduced. For now, just know that red will indicate 'rook'. In this section, good ideas – to shortly be rejected because they do not lead to chess-like play – will be shown in black; only ultimately correct solutions will be in color.⁵

The rook will be our guide, our paradigm trail blazer. We join our hero in his finest moment, the endgame.

⁵ I'm frankly curious if this pedagogical tactic will travel.

Obstacle #1 – Demise of the Endgame

Figure 1 shows the classic endgame of a single rook and king against a lone king. In 2D, the rook's move cuts the board into 4 disjoint regions (a). The opposing king will find itself trapped in one of them. The White king and rook work together to shrink the *area* of the region confining the Black king until mate is achieved.

However, in 3D, even with access to the full vertical extent of the board, a rook's straight-line move fails to cut the 3D board into disjoint regions and the opposing king is free to move to any tile not directly under attack (b). Mate is not possible with a single rook, nor with both rooks, nor even with three (which is more than either player even starts with).

This is not the only endgame to crumble under the weight of three dimensions; most of them no longer work at all, with the catastrophic result that games are unwinnable without a vast material advantage. Games between players of equal skill therefore end in stalemate; boring – and again, not like chess.

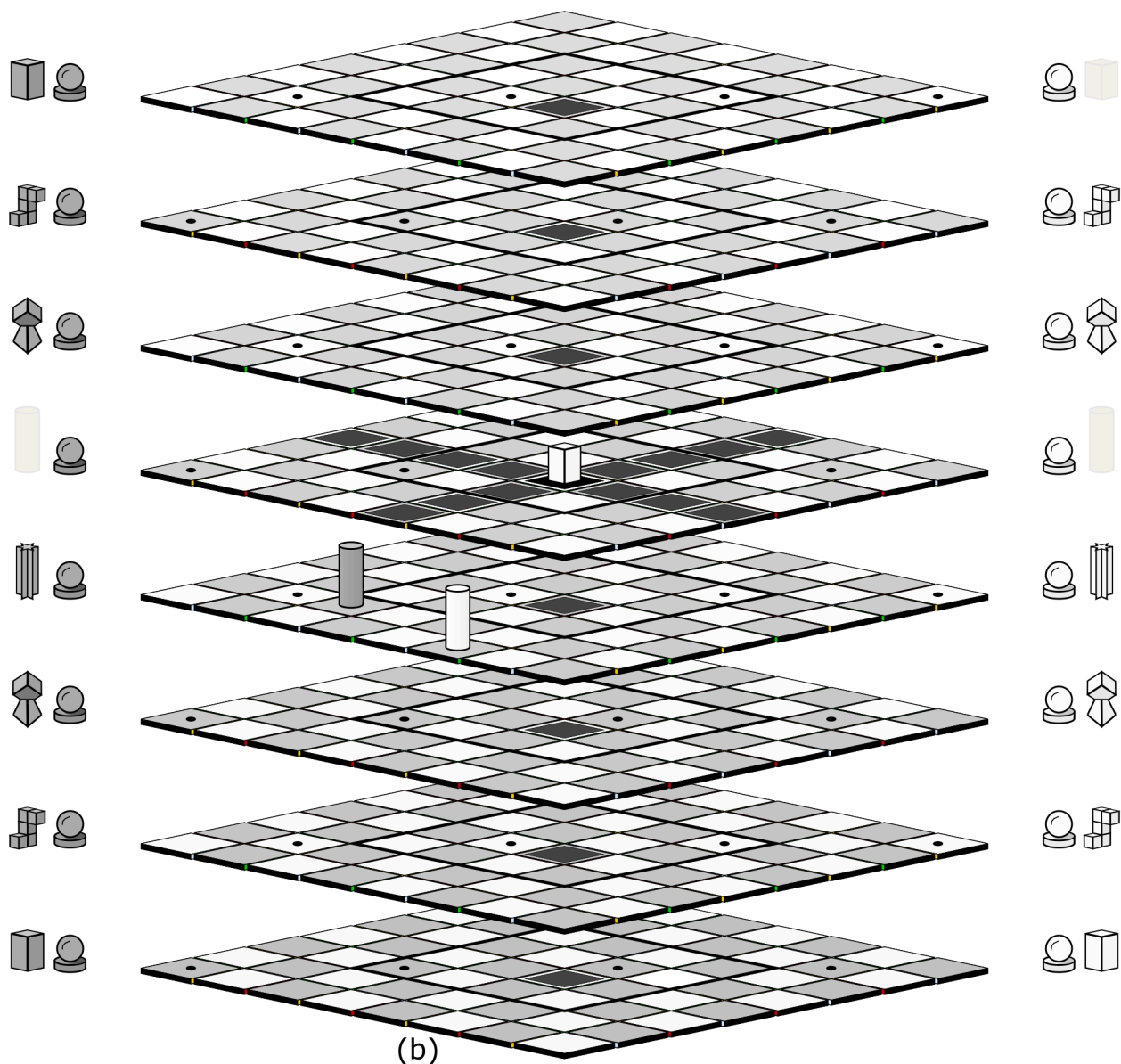
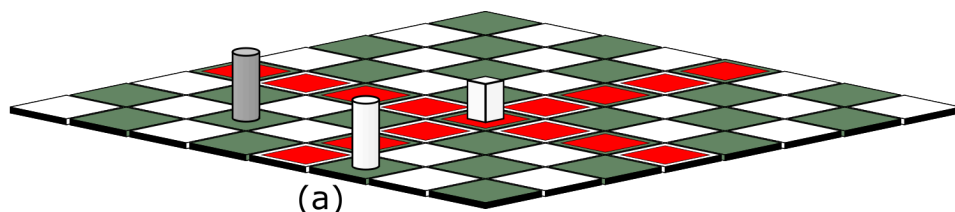


Figure 1: Single Rook Mate Endgame Cannot be Done with Straight-line Moves

In 2D (a), a straight-line rook move cuts the board, so a rook and a king (White) can checkmate a lone king (Black) – the most basic endgame scenario. In 3D (b), a straight-line rook move does *not* cut the board and so the mate algorithm fails – doesn't feel like chess. What kind of a move would cut the board?

What kind of a move would cut the board into disjoint regions, regions that would confine the lone king, regions that could be shrunk to zero, allowing a successful mate of the opposing king? To ask the question is to very nearly answer it, but let's sneak up on it anyway. In 2D, the major pieces that advance (all but the knight) do so along rays (directed line segments of unit length), that change either one coordinate or two. The rook advances along a ray that changes only *one* coordinate; thus, it moves through the sides that connect adjacent squares. The bishop advances along a ray that changes *two* coordinates; thus, it moves through the corners that connect adjacent squares. Call the first a *side ray*, and the second a *corner ray*. The queen may advance along either a side ray or a corner ray.

Because a side ray changes only one coordinate, the rook can reach *every* square on the board, but because the corner ray changes two coordinates, the bishop can reach only *half* the squares on the board.

In 2D, pieces move in straight lines, along a *single* ray, and thus they move in a one-dimensional space; one dimension less than the 2D board. If in 3D, pieces moved in planes (flat surfaces), along *two* rays, they would be moving in a two-dimensional space: one dimension less than the 3D board. Call this a *planar move*.

In 2D, there are *two* types of rays; *side rays* that connect squares through their sides and change only *one* coordinate, and *corner rays* that connect squares through their corners and change all *two* coordinates. One type of ray for each type of neighbor; side and corner.

In 3D, there are *three* types of rays; *face rays* that connect cubes through their faces and change only *one* coordinate, *edge rays* that connect cubes through their edges and change *two* coordinates, and *vertex rays* that connect cubes through their vertices and change all *three* coordinates. One type of ray for each type of neighbor; face, edge, and vertex.

Planar Moves

Figure 2 shows exactly the same state of the boards as the previous figure but this time with planar moves on the 3D board rather than straight-line moves. This changes everything. Instead of the rook moving in three *orthogonal lines* it is now allowed to move in three *orthogonal planes*.

These three planes slice the board into disjoint regions (now eight instead of four). Under this move, the opposing king will find itself trapped in one of them. The White king and rook can now work together to shrink the *volume* of the region confining the Black king until mate is achieved. Indeed, it is exactly the same algorithm as in 2D, just a few extra moves to confine the king against the extra dimension. See “Chapter 19: Endgames” for the details.

The previous straight-line moves are indicated with dark inserts in the red diamonds, they indicate the face rays. Each line (two rays in opposite directions) is the intersection of two planes. The rook is moving in a space only *one* dimension less than the board, same as in 2D. In contrast, the straight-line move on a 3D board was moving in a space *two* dimensions less than the board (previous figure). The planar move is the critical breakthrough that allows play in three dimensions to feel like chess.

By now the answer should start to feel inevitable; pieces which moved in *lines* in 2D, should move in *planes* in 3D. In the case of the rook this means three *orthogonal* planes; one horizontal, two vertical.

Obstacle #2 – Blocking

Given a pair of rays, it is natural to assume that a piece advances first along one ray then along the other. Two different rays define a plane. A pair of rays is the very foundation of the planar move. For instance, one might imagine a rook advancing along one face ray out N tiles, then advancing M tiles along another face ray at right angles. The rook is advancing along each ray through the faces connecting adjoining cubes, changing just one coordinate at a time. But there is an ambiguity because the rays may be followed in either order. Is the move MxN or is it NxM?

Figure 3 shows the ambiguity. There are two paths; White has a dual attack on a single piece, and Black cannot block both in a single move. In 2D, that can't happen, an attack always comes from *only one direction*. Because of symmetry, both paths must be allowed; blocking is busted; there go the chess gambits, there goes the midgame. The straight-line move didn't feel like chess, and the *dual path move* doesn't either, yet we were led to it inexorably by the requirement to have the pieces move in planes. Only planar moves can save the endgame. A pair of rays define a plane, but they also imply dual path attacks which cannot be blocked in a single move. Our quest, for the 'one right rule set', requires that both the endgame and the midgame feel like chess.

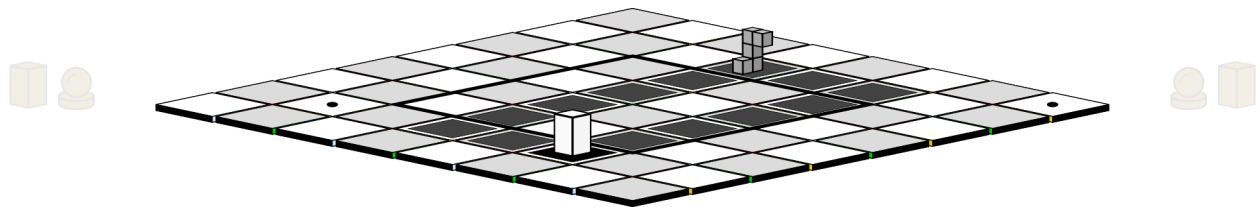


Figure 3: Rook's Dual Path Attack on a Knight – Rejected

If a ray pair defines a plane, then a planar move might be regarded as two straight-line moves at once; first down one ray, then down the other. However, symmetry implies both traversal orders must be allowed. That means an attack occurs along *two* paths; the rook attacks both down the 3x6 path and down the 6x3 path. Black cannot defend in a single move. Blocking busted; there go the gambits, there goes the midgame – doesn't feel like chess.

We are being humbled; the paradigm is fighting back. Perhaps there are no rule sets that can reproduce the feel of chess in three dimensions. Perhaps, sigh, chess is exclusively a two-dimensional game. The quest is blocked, by blocking of all things, how ironic.

We were successful in rising to the challenge of figuring out what kind of move cut the board (a planar move). Now we face a new question, "What kind of a move can make a planar attack,

but which can be blocked in a single move by the opponent?” How does a dual path attack get reduced to an attack, “from only one direction?”

One possibility is to declare that *both* paths must be unblocked for the piece to be under attack. This allows the attack to be blocked by placing a third piece on *either* path. While counter-intuitive, it does at first glance appear to solve the blocking problem.

STAB (Simultaneous Threat And Barter)

While the rules may require a process of discovery, naming things is a process of invention. It is useful to introduce the term *cover*, to indicate either attacking an opponent’s piece or defending one’s own piece. *STAB* is the measure of cover. The ideal *STAB* value is how many pieces any one piece can cover from the center of an empty board. In 2D, that’s 4 for the rook and bishop, and 8 for the queen, king, and knight.

‘Simultaneous Threat and Defense’ would be more accurate but leads to a meaningless acronym, ‘stad,’ while ‘Simultaneous Attack and Defend’ is even worse.⁶ If ‘barter’ is regarded as an exchange, “if you take mine, I’ll take yours,” then *STAB* works out very nicely.

The problem is that the *STAB* value under the dual path rule is obscenely high.

Figure 4 shows the problem. The rook can circumvent closer pieces and attack ones farther away. The 13 ghost rooks show the potential, but for clarity only one dual path attack is shown; that’s 13 pieces per plane. If the planar move is envisioned as a dual path, even with the blocking rule that obstructing *either* path blocks the attack, such an interpretation results in an astonishing *STAB* factor of 39! The rook is able to circumvent closer pieces and reach ones ‘behind’ it. A single piece could cover all the other pieces on the board – it won’t feel like chess.

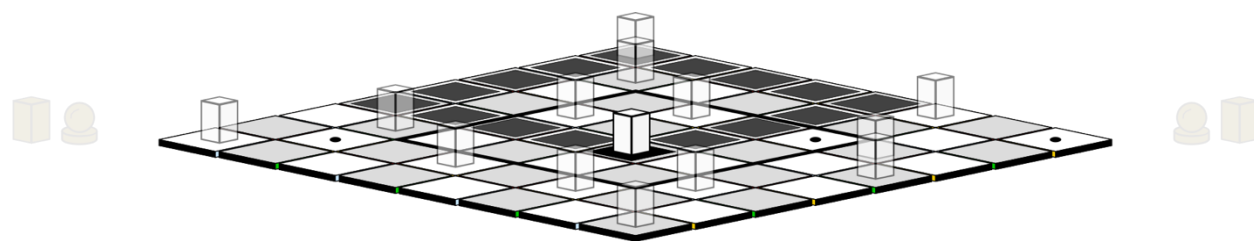


Figure 4: Rook’s Outrageous STAB of 39 for Dual Path Move - Rejected

A rook allowed to move along dual paths can attack or defend 13 other pieces in each plane. Shown is one rook near the center of a horizontal plane. For clarity only one dual path is highlighted. Similar diagrams would apply for both vertical planes (not shown). The ghost rooks show the moves legal under the dual path rule.

While initially promising, the dual path rule fails to feel like chess. However, this option does have an upside, it introduces the concept of nonlocality; the mere existence of something *over*

⁶ Technically, it’s just SAD.

there (on the other path) determines if a piece's presence *over here* (on this path) is causal or not. In the jargon of brainstorming it is a transitional concept; a 'bad' idea that leads to a 'good' idea.

Nonlocality

We begin by backing up.

How does a piece in 2D really move? Does it advance until it runs into something? It certainly feels that way when we pick up a physical piece and move it from one square to another. Although, come to think of it, we do generally pick – it – up, into the third dimension, rather than sliding it along the surface of the board proper.

Is that the only way of describing a move? Recall that a piece starts in one tile, ends up in another, perhaps many tiles away. It was *here*, now it's *there*. Was it really, *ever*, anywhere in between? (Look at a listing of the moves to a 2D chess game; those intervening squares are nowhere to be found.)

The physical board and pieces are not the game; they are certainly not the rules. They are only conventionalizations that help our finite minds enforce the rules. Only the rules are the game. A mathematician would call the game a *formal system*.ⁱⁱⁱ

In between *here* and *there*, were some small number of tiles. Anything unique about them? We are talking just about pieces with unlimited range, pieces that can be blocked (the knight is another matter). Back to the in-between tiles; what do they have in common?

They have empty in common.⁷ The set of in-between tiles contain no pieces, not a single one. The set happens to have, in 2D, the geometrical form of a line segment. A line segment is in a space *one dimension less* than a 2D board. A 'plane segment' (whatever that means) would be in a space *one dimension less* than a 3D board.

2D chess confronts us with a *conceptual degeneracy*: does a piece advance until it runs into another piece, or does it just advance until some growing set of tiles (along a single ray) ceases to be empty? In the latter case, call the set of tiles which must be empty an *advancement manifold*.

A first guess at an answer is shown in Figure 5.

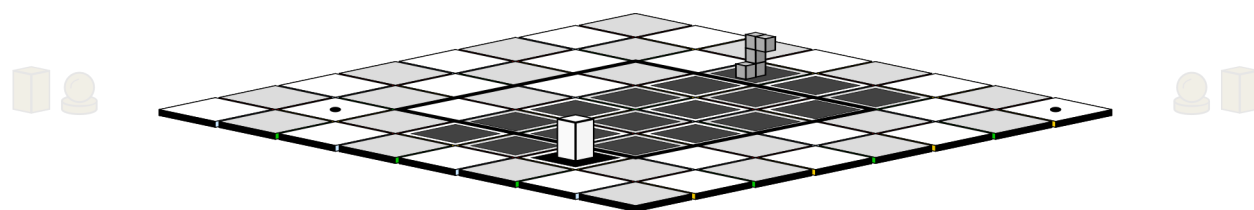


Figure 5: The Rectangular Advancement Manifold (RAM) – Rejected

White's rook is attacking the Black knight via a *rectangular advancement manifold* (RAM for short). There can be no other pieces in the 18 tiles that make up this RAM. Black can block this attack with a single move by moving another piece onto *any* of the 16 empty tiles marked in black.

⁷ You don't know how badly I wanted to say, "they have *nothing* in common."

In 2D, the advancement manifold is a line segment, in 3D, it is, perhaps, a rectangle, and the $M \times N$ rectangle must be empty of any other pieces for the move to be legal.

Here is a blockable planar attack; gambits restored, midgame saved, all without sacrificing the endgame. Not too bad, not too bad at all.

So why is Figure 5 in black and white?

Obstacle #3 – Openings

The RAM rule is easy to visualize, it is easy to present and describe, and it is even easy to program. But this paradigm barrier has another punch to throw. Does an $M \times N$ advancement manifold grow by M or by N ? In the above example of the growing advancement manifold, were tiles added three-at-a-time 6 times, or were tiles added six-at-a-time 3 times? The ambiguity stinks.

A further smell is that moves with skinny rectangular manifolds will be harder to block than moves with fat rectangular manifolds. The fattest rectangular manifolds will actually be square, so the rule isn't very isotropic either. But it gets worse.

The worst-case STAB value is still 39.

Figure 6 shows an example, broken into 3 parts for enhanced clarity.

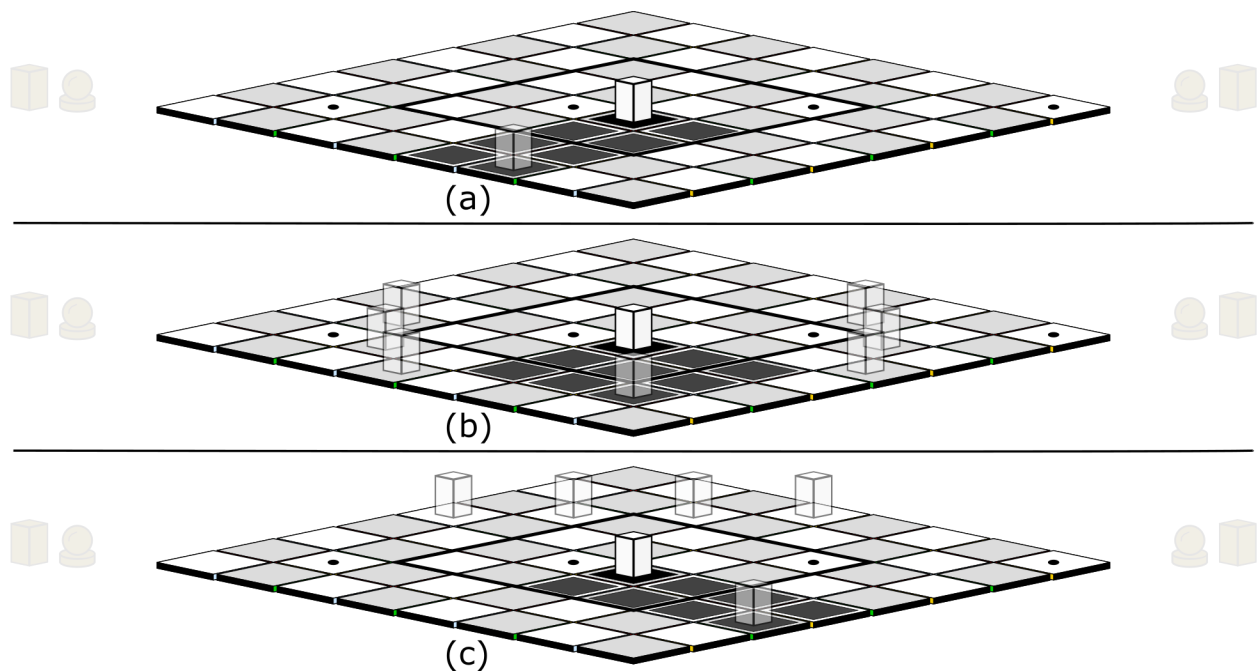


Figure 6: RAM STAB – Rejected

Under the RAM rule, the rook STAB value is still 39. Ghost rooks show the 13 locations that can be simultaneously covered by the rook under the RAM rule in one plane. None are straight-line moves, so the other two planes also support 13 targets. For clarity, only 3 rectangular advancement manifolds are shown (horizontal plane); ghost rooks are segregated for better visibility (a) – (c).

The starting lineup for 3D chess has not been derived yet, but the cover kind of gives it away. Note that a $1 \times N$ advancement manifold must be allowed, else straight-line moves would be illegal. Given that starting lineup, the side effect of straight-line moves is devastating; under both the dual path rule and the RAM rule, the rook, bishop, and queen can all move before a single pawn has advanced. There go the openings – doesn't feel like chess.

The advancement Square

The challenge remains; how to specify a planar move such that it simultaneously preserves the standard endgames, the midgame, and the strategic principles of openings.

Perhaps we can tease some further clues out of 2D chess. If the actual rule of motion is that the 2D advancement manifold can grow only as long as some set of tiles is empty, then the algorithm for legal motion must be to add one tile at a time and test if it's empty.

How should a 3D advancement manifold grow? If a line segment grows one tile at a time, then a 'plane segment' should grow one *perimeter* at a time. Call this an *advancement square*.

Figure 7 shows a series of snapshots of the rook's expanding advancement square for the same rook attack on the knight as in Figure 3.

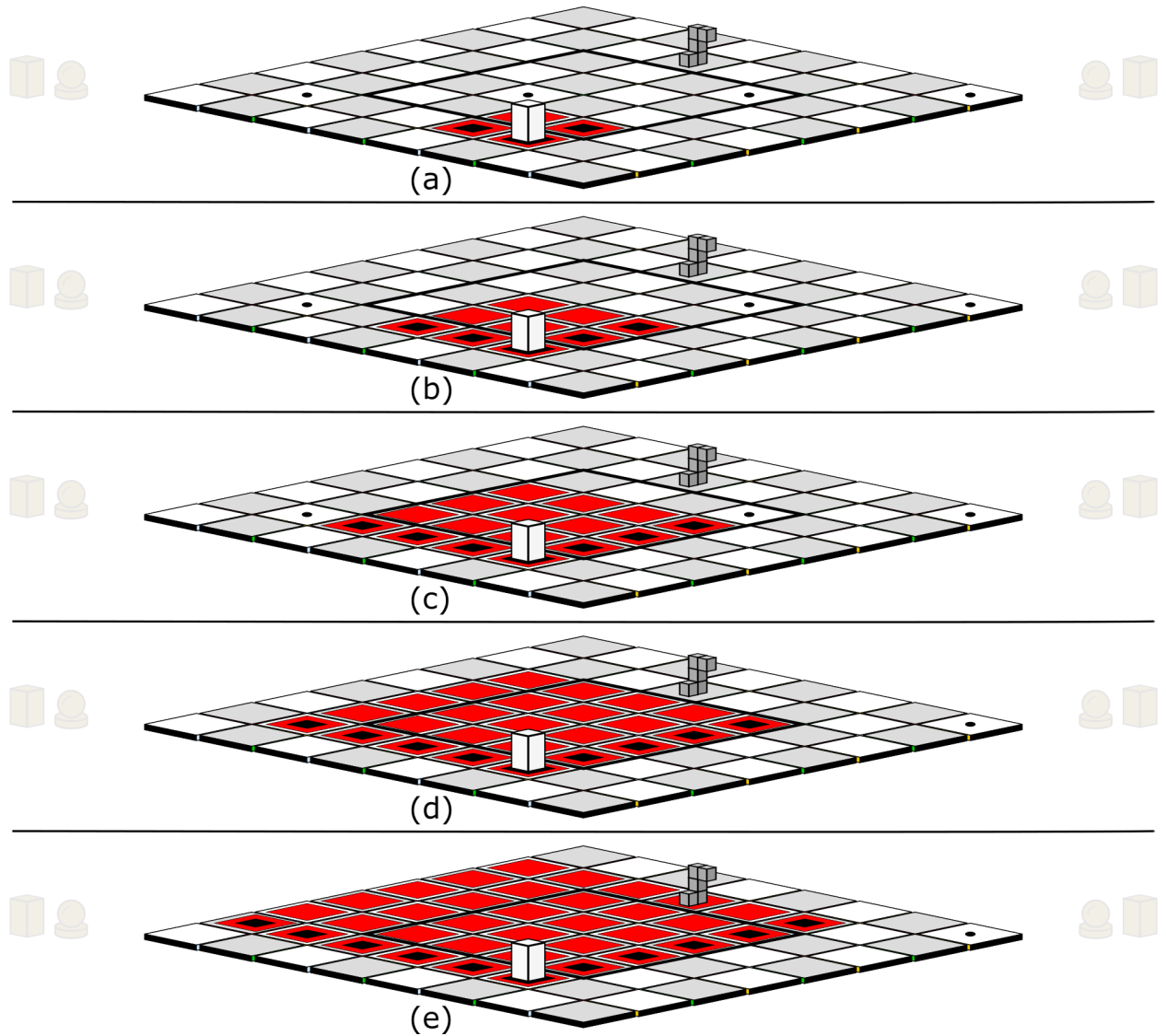


Figure 7: The Advancement Square

A series of 5 snapshots (a) – (e) showing each new perimeter of the 6x6 advancement square which must be empty of any other pieces for the White rook to be attacking the Black knight. Each perimeter is two tiles larger than the previous one (3, 5, 7, 9, 11). The line of red diamonds with black inserts represent the two rays (the inner edges of the advancement square). The presence of the knight on the perimeter (outer edges of the advancement square) prevents the rook from moving to those tiles (or beyond); its only legal move on that perimeter is to capture the knight. All the diamonds are slightly smaller than a tile so the underlying tile's color (black or white) can peek through.

The key is to constrain the advancement manifold to be *square*, 2×2 , 3×3 , ... $N \times N$. To move a 3D rook first requires the specification of a plane (one of three) and a quadrant (one of four). The rook can now attack only 4 pieces per plane, for a STAB value of 12, much more reasonable.

Figure 8 shows a series of snapshots of the rook's expanding advancement square in a vertical plane, in an upward quadrant toward Black. In this case, the advancement square shows that Black's king is in check.

Note that parts of an advancement square may extend off the board, as in this example. In the case of a rook, it leaves a rectangular shaped manifold behind (3x4). Tiles off the board will be occasionally shown when it helps visualize a move; however, the tiles in any part of an advancement square off the board are regarded as unoccupied and thus cannot block the move. Obviously, a piece cannot move off the board.

Some conventions are evident in how advancement squares are going to be presented in this work. Colored diamonds just smaller than the board tiles will be used as *decorators* to indicate that a tile is on some special part of the advancement square, something of tactical significance. Each type of piece will have its own color. Using decorators a little smaller than the tiles allows the color of the underlying tile (black or white) to peek through.

The number of decorators is significant, so to avoid overwhelming the reader, they will be introduced as needed. A summary of them is provided in Appendix II for reference. For now, only two decorators will be used. A plain decorator indicates a tile in the body of the advancement square. One with a solid black diamond insert indicates a tile along one of the two rays that specify the plane and quadrant of the move (the *inner two edges* of the advancement square). The perimeter is the *outer two edges* of the advancement square.

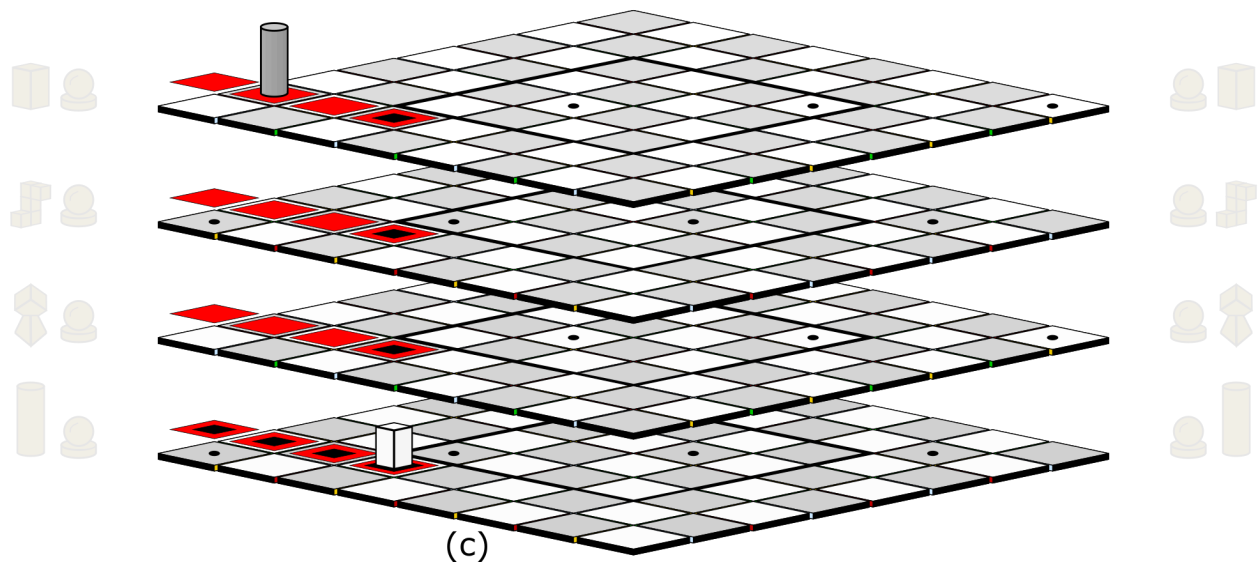
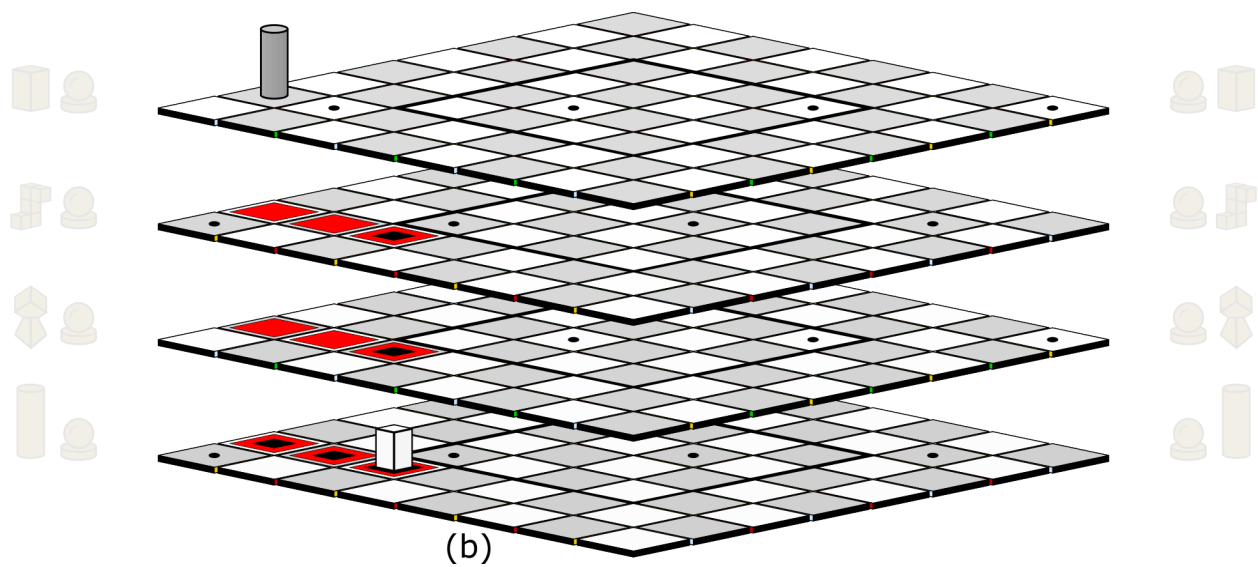
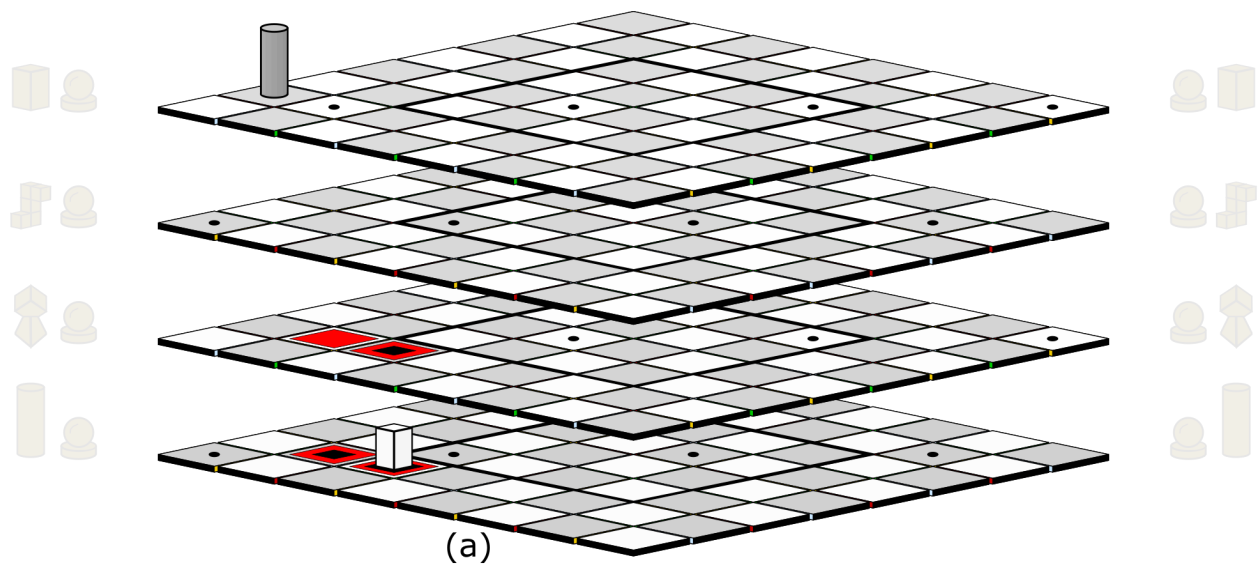
These conventions should be evident in the previous two figures which show a 6x6 advancement square in a horizontal plane, and a 4x4 advancement square in a vertical plane.

The king is covered (attacked, checked actually) by White's rook.

The piece shadows on either side of the board represent the *tray*; where captured pieces go. These shadows serve a dual purpose; they also label each level – king-rook, queen-bishop, whatever.

Figure 8: Three Snapshots Showing King in Check by Rook in Left Vertical Plane

(Facing page.) Three snapshots (a) – (c) proving the Black king is in check by the White rook via a left vertical advancement square. Note how part of the advancement square extends off the board.



Each plane is divided into four *quadrants* separated by face rays. Figure 9 shows the four quadrants in a horizontal plane. They are shown in opposite pairs for clarity. The ghost rooks indicate potential *quadrant moves* at particular perimeters of various sizes. Note that the diamond inserts at the ends of the perimeter are not red, they are black. These represent *linear moves* to be clarified next.

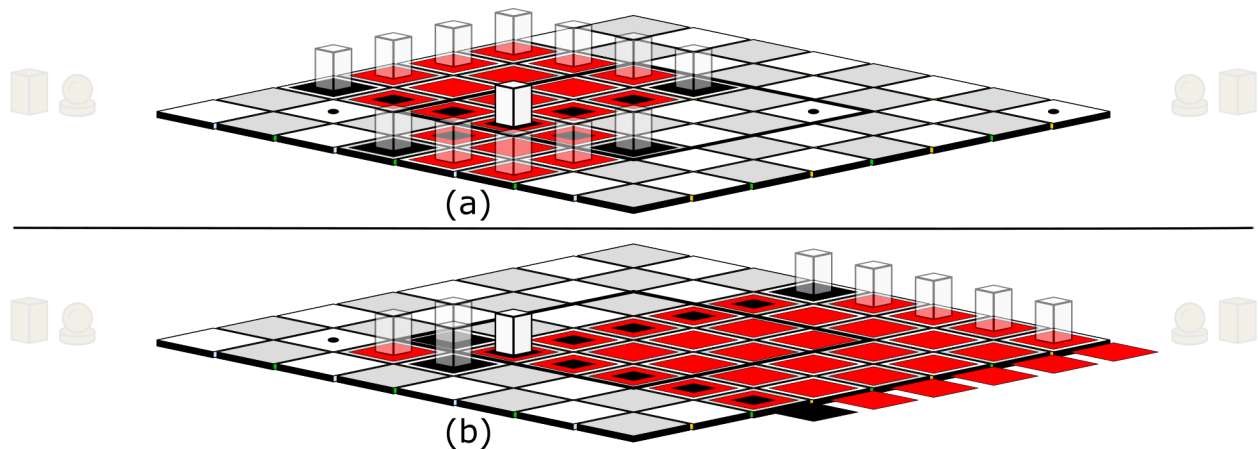


Figure 9: The Four Quadrants of the Rook Move

The four quadrants are shown in two pairs for clarity. A 3x3 and a 4x4 advancement square (a), and a 2x2 and a 6x6 advancement square (b). The tiles which the rook can move to on the respective perimeters are shown with ghost rooks. Note that the 6x6 advancement square extends off the board, so the ghost rooks on those tiles are not shown. Note also the subtle implication that linear moves will need to be dealt with as special cases, for the tiles under them are black not red.

Advancement squares expand into quadrants; what then of linear moves?

Obstacle #4 – Linear Moves

A linear move in 3D presents two challenges, which will be referred to as *quadrant ambiguity* and *plane ambiguity*. Recall that two planes intersect in a line. A line of tiles along a ray form the intersection between two planes but also form a common boundary between two adjacent quadrants within *each* of those planes. It is ambiguous in which *quadrant* a linear move advances, and it is ambiguous in which *plane* a linear move advances; all, either, neither, both?

There are a number of possibilities here;

- let a linear move be legal if *any* quadrant is clear
- allow only if *all* the quadrants are clear
- group in pairs by plane and allow the move if *either pair* is clear

The first option can be easily dispensed with. In the starting lineup, the major pieces are supposed to be blocked by the requirement of the advancement square, but as we have seen, it is possible for the advancement square to extend off the board. If a linear move is allowed when *any*

quadrant is clear, then the starting lineup only blocks two of the four quadrants. For the rook, the pawn blocks motion in the horizontal plane for the quadrant extending inward, but it does not block the quadrant that extends outward off the board. Similarly, the knight blocks motion in each vertical plane only for the quadrant extending downward, but it doesn't block the quadrants that extends upward off the board. Under the *any* rule, the rook, bishop and queen can all move before a single pawn moves. We've seen this dance before; there go chess openings.

The second option can also be dispensed with. If *all* quadrants have to be clear, then it would be four times easier to block a linear move as a quadrant move; there goes the ideal of isotropic moves. Also, in 2D chess, the lines are independent, what happens in one has no direct effect on the other. Manifold independence (line, plane) should be an invariant which is honored. Thus, in 3D chess, the planes should be independent; what happens in one should have no direct effect on the others.

The Advancement Rectangle

This leaves only option three; motion is allowed if *either pair* of adjacent quadrants are clear, one plane or the other may be blocked, but as long as both planes are not blocked, the move is legal. However, both quadrants in at least one plane must be clear. A linear move may occur in only one plane, or it may occur in both, depending on other pieces. This means that the advancement manifold for a linear move is actually an *advancement rectangle*. This is not to be confused with the debunked rectangular advancement region (RAM) which is part of why such a clumsy name was chosen for that concept; it was intended to avoid later confusion.

While it is almost twice as likely to block a linear move in a given plane, it is twice as likely to be unblocked in at least one of the planes; that restores the ideal of isotropic moves.

Figure 10 shows three snapshots of how the advancement rectangle grows with each new perimeter in just one plane, in this case, the horizontal plane. Note that the perimeter for a linear move grows by 5, 9, 13, ... tiles, twice as fast as for an advancement square. If any of these tiles are occupied, then an advance to that perimeter is blocked.

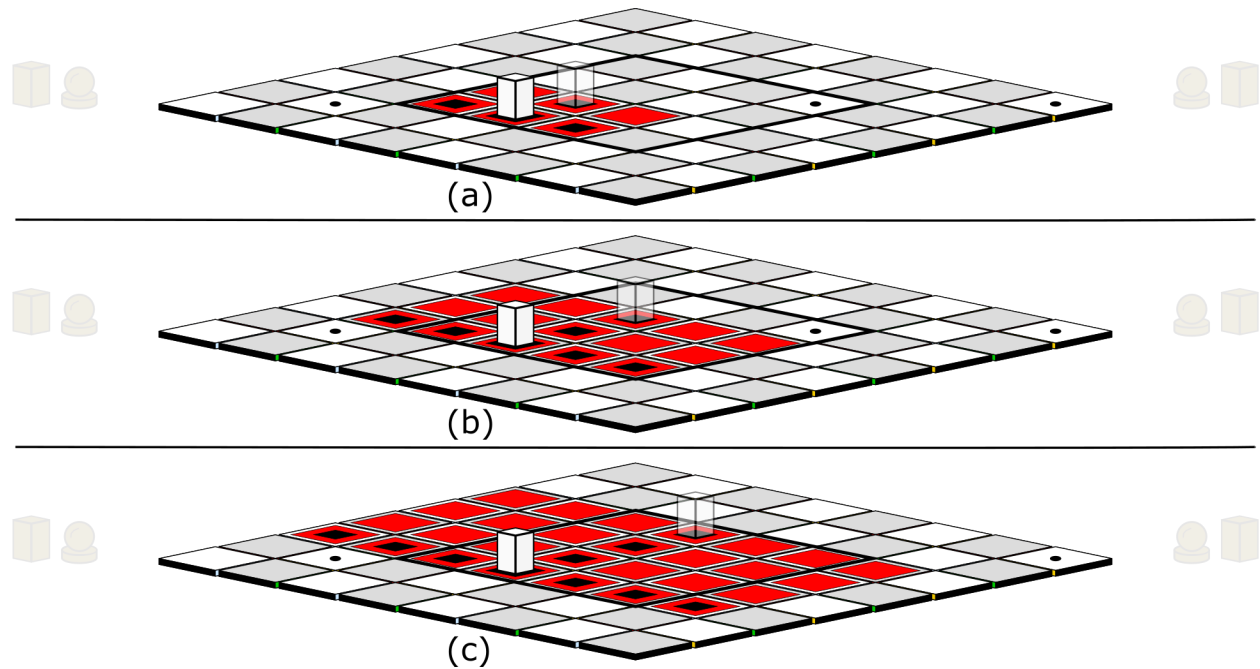


Figure 10: Rook's Linear Move

In each plane, a linear move is ambiguous in which quadrant it advances in, so both advancement squares must be empty for the move to be unblocked in that plane. Three snapshots (a) – (c) of the rook's *advancement rectangle* are shown for the horizontal plane.

Figure 11 shows two complete advancement rectangles for a linear move in the intersection of a horizontal and a vertical plane. Advance is eventually blocked by another piece in each plane; the knight blocks the vertical plane limiting the rook's move to perimeter 2 (corresponding to two 3x3 advancement squares, i.e., a 3x5 advancement rectangle), and the pawn blocks the horizontal plane limiting the rook's move to perimeter 4 (corresponding to two 5x5 advancement squares, i.e., a 5x9 advancement rectangle). An advancement rectangle 9 tiles long will obviously extend off the board. Because of their larger size, it is quite common for advancement rectangles to do so. The tile with the ghost rook (transparent) can only be reached in the horizontal plane, two of the three previous in-line tiles were also reachable in the vertical plane.⁸

The advancement square has a STAB value much closer to 2D; from a low of 4 for the maximum number of simultaneous linear attacks to a high of 12 for the maximum number of simultaneous quadrant attacks.

⁸ Interesting, linear moves can be *polarized*.

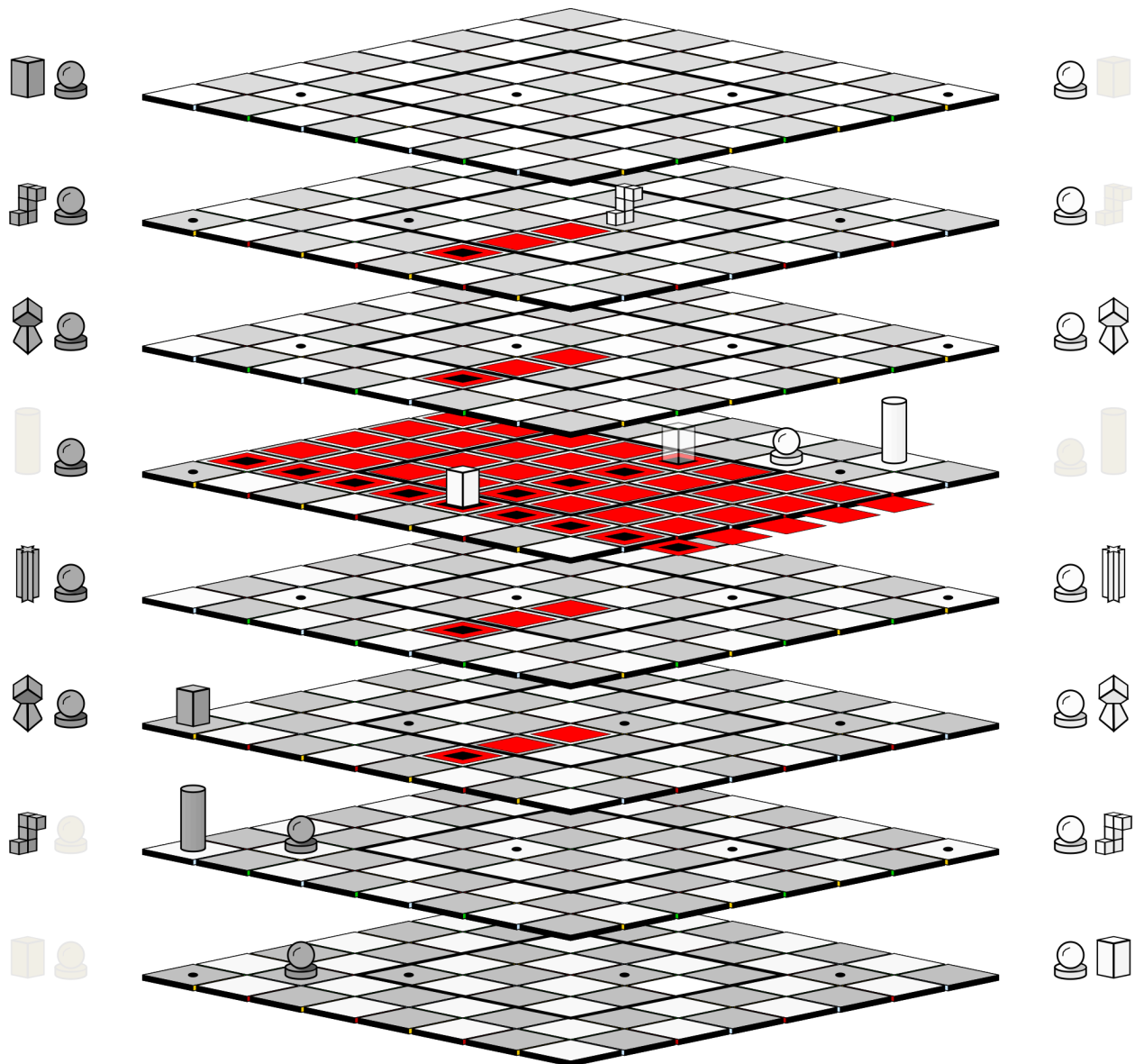


Figure 11: Linear Rook Move in Horizontal and Vertical Planes

The two advancement rectangles for a linear rook move are shown. The vertical one is blocked by the knight, the horizontal one blocked by the pawn. The tile containing the ghost rook (king level) can only be reached in the horizontal plane, the first two linear tiles are reachable in both planes.

Loss of Trajectory

The implication of an advancement manifold is that a piece does not have a *trajectory* from its source tile to its destination tile; there is no unique sequence of tiles which it traversed. This is a bit of a surprise, but the steps that led here are inescapable once the commitment is made to hold discovery ascendant over invention. Who would make up such a rule? It violates our everyday

experience, all the perceptual algorithms that evolution has built into us over millions of years, and even violates the early development of science, now⁹ referred to as classical physics.

In 2D chess, it was possible to pretend that a move had a definite trajectory, a uniquely specified sequence of tiles connecting the source tile with the destination tile. In 3D chess, the pretense of a trajectory is no longer tenable. The loss of trajectory violates common sense in deeply profound ways.¹⁰

In classical physics, it was possible to pretend that an object in motion had a definite path, a specific and unique sequence of intermediate points, but according to quantum physics, it doesn't and never did. Neither was there ever a trajectory in 2D chess, but the physicality of the pieces, the very act of picking one up and moving it, deceived us. Paradigms are strong barriers.

Motion is not classical; it is fundamentally quantum. The perimeter of the advancement square is a portent of the concept of an advancing wave front. How did that happen? By a disciplined adherence to the discovery principle.

3D chess is an example of a game that is echoing fundamental principles of physical reality, even, to a tiny degree, the historical progression from a naïve classical view of motion to a more refined quantum view of motion. How very delightful – how very unexpected.

Had the loss of trajectory been recognized in classical chess long ago, then it might not have been such a scientific shock to discover that an electron in an excited atom is *first* in one orbital, *then* in another, but is *never, ever* anywhere in between.

“Oh,” we might have said, “it’s just like chess.”

The ancient Greeks’ love of philosophy lead to many inventive ideas on the ultimate nature of reality, but until the rise of the scientific method, little progress was made discovering the correct laws that truly reveal the nature of reality; discovery is ascendant over invention.

Our commonsense paradigm of motion is *classical*. Therefore, we typically invent games where pieces advance along well defined and unique trajectories, but adherence to the discovery principle reveals that trajectories are an *illusion*.

The trajectory paradigm, like all paradigms, seems *so opaque* on the front end, and so *bloody transparent* on the back end. It is one of the strong reasons to believe that the planar move, constrained by the advancement square, supports the hope that for 3D chess there just might be, *one right rule set*.

ⁱⁱⁱ Hofstadter, Douglas, *Gödel, Escher, Bach: An Eternal Golden Braid*, Vintage Books, New York, NY, 1979.

⁹ “fondly”

¹⁰ I have witnessed individuals led to the planar move reject it because they could not determine a workable trajectory.